Chapter 6 — Heat and Thermodynamics

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. First law of thermodynamics: (a) $Q = \Delta U - W$ (b) $W = Q + \Delta U$ (c) $Q = \Delta U + W$ (d) $\Delta U = QW$

2. Work at constant pressure: (a) $V\Delta P$ (b) $P\Delta V$ (c) P/V (d) $\Delta P/\Delta V$

3. In isothermal ideal gas: (a) $\Delta U = 0$ (b) $Q = 0$ (c) $W = 0$ (d) $P = 0$

4. In adiabatic process: (a) T constant (b) $\Delta U = 0$ (c) $Q = 0$ (d) P constant

5. Carnot efficiency: (a) $1 - T_1/T_2$ (b) T_1/T_2 (c) T_2/T_1 (d) $1 - T_2/T_1$

6. Refrigerator COP: (a) Q_1/W (b) W/Q_2 (c) Q_2/W (d) Q_1/Q_2

7. Entropy change: (a) $\Delta Q \times T$ (b) $\Delta Q + T$ (c) $\Delta Q/T$ (d) $T/\Delta Q$

8. In irreversible process, total entropy: (a) increases (b) is infinite (c) is zero (d) decreases

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why does the internal energy of an ideal gas depend only on temperature?

2. Why is the change in internal energy zero in an isothermal process?

3. Why does the temperature fall during adiabatic expansion?

4. Why does a refrigerator not violate the second law?

5. Why must kelvin temperatures be used in Carnot efficiency?

6. How is entropy related to the availability of energy for work?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) State the first law of thermodynamics and apply it to isothermal and adiabatic processes. **(4)**

(b) Show that the area under a P–V graph equals the work done. **(4)**

Q.4 (a) Describe the Carnot cycle and derive its efficiency. **(4)**

(b) Explain the working of a refrigerator and derive its coefficient of performance. **(4)**

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